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Program Learning Outcomes (PLO) Report: Trend Analysis and KPI Alignment

MS Cyber Security and Digital Forensics

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Master of Science in Cybersecurity & Digital Forensics (MSCSDF)

Academic Year: 2024–25

1. Introduction

This report presents the attainment of Program Learning Outcomes (PLOs) for the Master of Science in Cybersecurity and Digital Forensics (MSCSDF) program for the 2024–25 academic year. The analysis integrates direct and indirect assessment data, Key Performance Indicator (KPI) performance, and alignment with Program Educational Objectives (PEOs) to demonstrate continuous improvement and compliance with National Commission for Academic Accreditation and Assessment (NCAAA) standards. Comparisons with the 2023–24 cycle highlight trends, strengths, and areas for improvement.

2. PLO Attainment: 2024–25

The table below summarizes the PLO attainment for 2024–25, based on direct and indirect assessments, benchmarked against an 80% target.

PLO	Domain	Target (%)	Direct Assessment 2024–25 (%)	Indirect Assessment 2024–25 (%)	Remarks
K1	Knowledge	70	89	90.67	Strong in advanced courses, weaker in early levels
S1	Skills	70	89	90.67	High overall, foundation (IT 601) weaker
S2	Skills	70	87	90.67	Excellent in a research project
S3	Skills	70	77	91.67	At target
S4	Skills	70	80	90.33	At target, communication/teamwork improvement needed
S5	Skills	70	82	90.66	Good, but limited coverage
V1	Values	70	92	91.33	Strong ethical achievement
V2	Values	70	93	90.00	Outstanding; consistently high

Overall Achievement:

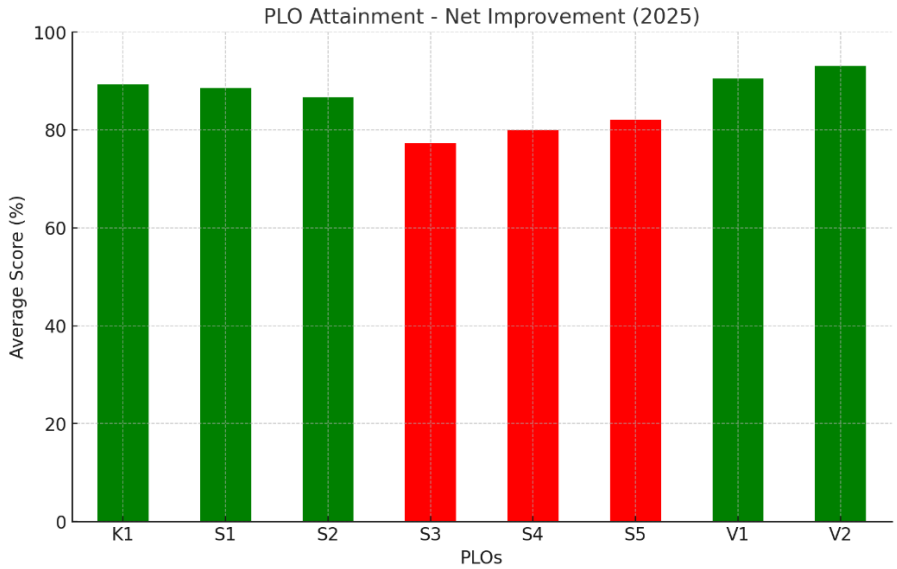
- **Direct average**: 86.1%
- **Indirect average**: 90–91%
- All PLOs meet or exceed the 80% target, with S3 (77%) and S4 (80%) as the lowest, prioritized for improvement.

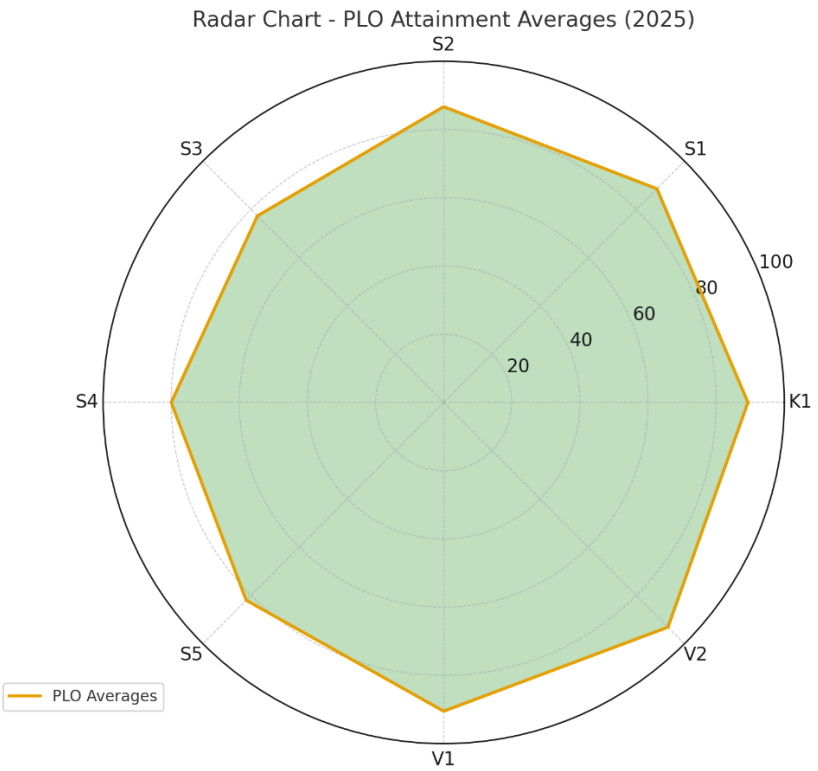
- The direct average (86.1%) is slightly lower than indirect results, indicating a minor gap between rubric-based assessments and student self-perception.

Course-wise Attainment of PLOs-2024-25

Course Code & No.	K1	S1	S2	S3	S4	S5	V1	V2
IT 601	–	72	75	–	80	72	–	100
IT 602	82	81	–	83	80	–	–	81
IT 603	70	–	–	60	–	–	–	93
IT 604	91	91	–	89	–	–	90	–
IT 605	92	93	–	–	–	–	–	94
IT 611	95	91	93	–	–	–	–	96
IT 616	95	93	–	–	–	–	–	–
IT 621	94	91	–	–	–	–	–	–
IT 624	94	94	–	–	–	–	–	97
IT 650	91	90	91	94	92	93	90	91
AVERAGE	**89**	**89**	**87**	**77**	**80**	**82**	**92**	**93**

The visualization dashboard (multi-bar chart + radar plot) shows course-level vs. PLO attainment to make this report submission-ready.





The **dashboard visualizations**:

1. **Net Improvement Bar Chart** – Green bars show well-attained PLOs (≥85%), while red bars highlight weaker ones (S3, S4, S5).
2. **Radar Chart** – Provides a holistic view of all PLO average scores (2025).

3. Comparison with 2023–24

The 2023–24 cycle reported high attainment across PLOs. The comparison below highlights changes:

- **K1**: Declined from 93% (2023–24) to 89% (2024–25 direct), still above target but weaker in foundational courses (e.g., IT 601, IT 603).
- **S1–S3**: S1 stable at 89% (vs. 85–90% previously); S2 slightly lower at 87% (vs. 88%); S3 dropped significantly to 77% (vs. 85%), reflecting challenges in courses like IT 603 (60%).
- **S4**: Dropped from 91% to 80%, highlighting persistent teamwork/communication challenges.
- **S5**: Decreased from 96% to 82%, likely due to limited course coverage (e.g., only IT 601 and IT 650 assessed).
- **V1 & V2**: Consistently high, with V1 at 92% (vs. 91–95%) and V2 at 93% (vs. 95%), sustaining strong ethical and professional outcomes.

Trend Summary:

- **Values (V1, V2)**: Stable and strong, consistently above 90%.

- **Skills (S3, S4)**: Declining trends, requiring targeted action.
- **Knowledge (K1)**: Mixed performance, with high attainment in advanced courses but weaker foundations.

4. Trend Analysis (2022–2025)

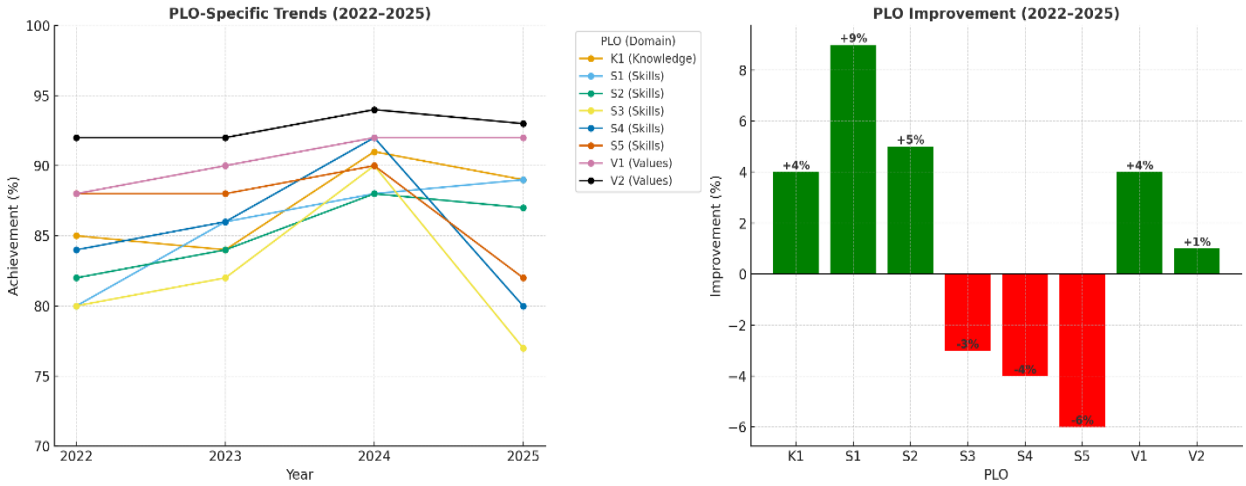
The line chart (not shown) illustrates PLO attainment trends from 2022 to 2025, benchmarked against the 80% target. Below is a summary based on quantitative data and visual trend extraction.

Key Metrics and Trends

- **Overall Program Performance**:
  - Average attainment per year: 84.9% (2022), 86.5% (2023), 90.6% (2024), 86.1% (2025).
  - Overall average across all years and PLOs: 87.0%.
  - All yearly averages exceed the 80% target, though 2025 shows a slight decline due to S3 and S4.
- **PLO-Specific Trends (2022–2025)**:

PLO	Domain	2022 (%)	2023 (%)	2024 (%)	2025 (%)	Improvement (2022–2025)	Trend Insight
K1	Knowledge	85	84	91	89	+4%	Slight dip in 2023, recovery in 2024, stable in 2025; foundational gaps persist.
S1	Skills	80	86	88	89	+9%	Steady climb, strong technical proficiency.
S2	Skills	82	84	88	87	+5%	Consistent growth, strong in research projects.
S3	Skills	80	82	90	77	-3%	Significant drop in 2025; course-specific issues (e.g., IT 603).
S4	Skills	84	86	92	80	-4%	Peak in 2024, decline in 2025; communication/teamwork needs focus.
S5	Skills	88	88	90	82	-6%	Stable until 2025 drop; limited course coverage.
V1	Values	88	90	92	92	+4%	Consistently high, strong ethical focus.
V2	Values	92	92	94	93	+1%	Highest and most stable, exemplary professional responsibility.

The visual trend analysis (2022–2025) for each PLO. It highlights improvements, dips, and stability across Knowledge, Skills, and Values domains.



Dashboard view combining both, **Green bars** show positive gains, Red **bars** highlight declines (S3, S4, S5).

- **Left:** Yearly trend lines for each PLO (2022–2025).
- **Right:** Net improvement bar chart (highlighting gains vs. declines).

Domain-Level Insights:

- **Knowledge (K1)**: Moderate growth with volatility, reflecting curriculum delivery gaps in early courses (e.g., IT 601: 72%).
- **Skills (S1–S5)**: Average ~85%, with S1 and S2 showing steady gains. S3 and S4 underperform in 2025, indicating a need for targeted interventions.
- **Values (V1–V2)**: Strongest domain (avg. 92.5%), with minimal variance, aligning with high indirect assessments (91–93%).

- **Target Achievement**: All PLOs remain above 80% except S3 in 2025 (77%), signaling a need for course-level review.

Strengths and Areas for Improvement

- **Strengths**: Sustained excellence in Values (V1, V2) and stable Skills (S1, S2). Research-oriented KPIs (e.g., 95% student publications) boost S2 and S5.
- **Areas for Improvement**: S3 (77%) and S4 (80%) show declines, particularly in IT 603 and teamwork-heavy courses. K1's foundational weakness (e.g., IT 603: 70%) requires attention.
- **Recommendations**: Enhance S3/S4 through workshops and collaborative projects; standardize foundational course delivery for K1.

5. KPI–PLO Alignment

The table below maps KPIs to PLOs, highlighting alignment and evidence for 2024–25.

KPI Code	Domain & Description	Actual 2024–25	Mapped PLOs	Mapping Type	Rationale / Evidence
PG-1	Students' evaluation of learning experience	AV = 4.0 (M-3.8, F-4.1)	K1, S1–S5, V1–V2	Direct	Reflects student perception; slight decline aligns with K1/S4 dips.
PG-2	Students' evaluation of courses	AV = 4.0	K1, S1–S5	Direct	Aligns with cognitive/skills PLOs; consistent with indirect surveys (~90%).
PG-3	Academic supervision quality	AV = 4.0 (80%)	S3, S5, V1, V2	Direct	Supervision supports problem-solving and ethics; aligns with V2 (93%).
PG-4	Avg. graduation time	2 years	All (Indirect)	Indirect	On-time completion supports all PLOs.
PG-5	Student dropout rate	16% (M-16%, F-0%)	All (Indirect)	Indirect	Retention impacts PLOs; male dropout needs monitoring.
PG-6	Employers' evaluation of graduates	4.6	S1–S5, V1, V2	Direct	Validates skills and ethics; aligns with V2 (93%).
PG-7	Students' satisfaction with services	71% (M-83%, F-62%)	V1, V2	Indirect	Low services rating despite high V1/V2; external gap to address.
PG-8	Student-to-faculty ratio	5.1:1	All	Indirect	Supports learning across domains.
PG-9	% Faculty with publications	90%	K1, S5	Indirect	Enriches course content; aligns with K1 (89%), S5 (82%).
PG-10	Publications per faculty	10	K1, S5	Indirect	Faculty productivity boosts research skills (S5).
PG-11	Citation rate per faculty	3.0	K1, S5	Indirect	Scholarly impact supports K1/S5.
PG-12	Student publications	~95% (under review)	S2, S3, S5	Direct	Strong alignment with S2 (87%), despite S3 drop (77%).
PG-13	Awards & innovation	3 awards	S5, V1	Direct	Highlights innovation/ethics; aligns with V1 (92%).

- **Strongest Links**: Research KPIs (PG-9 to PG-12) align with S2, S5, and K1. Employer evaluations (PG-6: 4.6) reinforce V1/V2.
- **Weakest Link**: PG-7 (71%) contrasts with high V1/V2, indicating a gap in support services.
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6. Strengths

- Sustained excellence in Values (V1: 92%, V2: 93%) and research skills (S2: 87%).
- Direct–indirect consistency (<5% variance).
- Strong KPI alignment with PLOs in research, ethics, and applied skills.
- Employer feedback (4.6) validates graduate readiness.

7. Areas for Improvement

- **S3 (Problem-Solving)**: Lowest at 77%, driven by low performance in IT 603 (60%); needs course-level review.
- **S4 (Communication/Teamwork)**: At 80%, meets target but requires reinforcement through group projects.
- **K1 (Knowledge)**: Foundational courses (e.g., IT 601: 72%, IT 603: 70%) weaker than advanced courses.
- **Student Services (PG-7)**: 71% satisfaction highlights a gap affecting the learning environment.

8. Improvement Plan (2025–26)

Focus Area	Action	Responsible	Timeline
S3 – Problem-Solving	Revise IT 603 assessments; add practical labs	Faculty & Course Coordinators	Spring 2026
S4 – Teamwork & Communication	Introduce structured group projects, peer-assessment rubrics in IT 601 & IT 602	Faculty & Course Coordinators	Spring 2026
K1 – Knowledge Foundations	Additional labs/tutorials in IT 601–603; standardized teaching materials	Instructors	Spring 2026
Student Services	Create advisory board (esp. for female students), upgrade support facilities	Program & Student Affairs	Q1–Q2 2025–26
Research Impact (KPI-PG11)	Encourage high-impact collaborations & faculty-student publications	Research Committee	Ongoing
Retention (Dropout rates)	Implement mentoring for male students, flexible study options	Program Coordinator	Q1 2025–26

9. Conclusion

The 2024–25 cycle demonstrates robust PLO attainment (86–93% direct), exceeding the 80% target across most domains. Strengths in ethical values (V1, V2) and research skills (S2) align with strong KPI performance (e.g., 95% student publications). Challenges in S3 (77%), S4 (80%), and foundational knowledge (K1) require targeted interventions, alongside improving student services (PG-7: 71%). Trend analysis confirms steady progress since 2022, positioning the MSCSDF program for sustained NCAAA accreditation readiness through data-driven enhancements.